Barbier's theorem for the Reuleaux triangle and the convex four-vertex theorem in Lean 4, through support functions

Josh Bald
Independent researcher, Ontario, Canada
jpbald93@gmail.com
ORCID: 0009-0002-1317-6489 *

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Abstract

We give machine-checked proofs, in Lean 4 with Mathlib, of two classical theorems on convex plane curves, both developed in one model: a curve is described by its support function h, and is traced as $\gamma_h(\theta) = (h(\theta) + i\,h'(\theta))\,e^{i\theta}$. The first is Barbier's theorem, in the form: if h has constant width w, the perimeter of γ_h, defined as $\int_0^{2\pi} |\gamma_h'|$, is πw. We prove it not only for twice continuously differentiable support functions but for support functions that are merely C^1, with h' differentiable and $h + h'' \geq 0$ outside a countable set and $h + h''$ integrable there. This covers the Reuleaux triangle, which we construct, show to satisfy the hypotheses, and show not to be twice continuously differentiable, so the smooth theorem does not reach it. For the Reuleaux triangle we also identify the perimeter with Mathlib's length and exhibit it as a Mathlib convex body of constant width. The second is the four-vertex theorem for ovals given by C^3 support functions with $h + h'' > 0$: the curvature is constant or has at least four alternating local extrema in a period. We tie each of our own definitions to a standard notion: the perimeter to Mathlib's length (total variation) — for smooth support functions, and separately for the Reuleaux triangle — via a lemma, not in Mathlib, that the length of a C^1 curve is finite and equals $\int |\gamma'|$; the support function to an actual convex set, whose support values and widths we compute; and our curvature to the textbook formula $(x'y'' - y'x'')/|\gamma'|^3$, for a curve we prove to be simple and closed. We do not prove the classical statements for arbitrary C^2 convex curves or arbitrary convex bodies of constant width: our Barbier and four-vertex theorems are about curves given by such support functions. An independent comparison with textbook statements confirms that the development covers them only in this narrower form, and that the area of the Reuleaux triangle and Kneser's non-convex four-vertex theorem are out of scope. All proofs use only Lean's three standard axioms. We know of no previous complete formal proof of either theorem, even in this restricted form, in any proof assistant, and no formalisation of the Reuleaux triangle.

1 Introduction

A compact convex set in the plane has *constant width* w if every pair of parallel support lines is at distance w. Barbier's theorem [1] says that every such set has perimeter πw: the circle of diameter

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w is far from the only example, but all of them have the circle's perimeter. The best-known non-circular example is the Reuleaux triangle, bounded by three circular arcs of radius w, each centred at the opposite corner; see [8] for the theory of constant width.

The four-vertex theorem [2, 3] says that the curvature of a simple closed smooth plane curve is either constant or has at least four local extrema. A standard proof for strictly convex curves goes through the support function and a Sturm–Hurwitz-type lemma: a continuous 2π-periodic function orthogonal to 1, cos and sin changes sign at least four times in a period, unless it vanishes; see [5, 4, 7] for the history and the converse.

Both theorems appear on the "1000+ theorems" list that Mathlib tracks, and neither has a formal proof there (Section 8). This paper gives complete Lean 4 proofs of both for curves given by support functions, in a common model, and says exactly how far each statement reaches; the classical statements for arbitrary convex curves and bodies are not proved.

Contributions.

1. **Barbier, with corners** (Theorem 6): a curve whose support function is C^1, 2π-periodic and of constant width w, with h' differentiable and $h + h'' \geq 0$ outside a countable set and $h + h''$ integrable (Definition 3), has perimeter $\int_0^{2\pi} |\gamma_h'| = \pi w$. The smooth version (Theorem 4) is a special case (Proposition 5).

2. **The Reuleaux triangle** (Theorem 8): an explicit support function, proved to satisfy the hypotheses of Theorem 6, to have perimeter πw, to have three corners pairwise at distance w and sides that are arcs of radius w about the opposite corner, and *not* to be twice continuously differentiable; and, with Mathlib's length in place of our own perimeter, that the length is πw (Theorem 8(f)).

3. **The four-vertex theorem for ovals given by C^3 support functions with $h + h'' > 0$** (Theorems 12 and 20), with the Sturm–Hurwitz lemma (Lemma 10).

4. **Bridges to standard notions** (Section 6): the length of a C^1 curve is finite and equals $\int |\gamma'|$ (Theorem 13); our perimeter is Mathlib's length for smooth support functions (Corollary 14) and, separately, for the Reuleaux triangle (Theorem 8(f)); the support function h is the support function of an explicit convex set of constant width w, packaged as a Mathlib convex body (Theorems 16 and 17); our curvature is the textbook signed curvature of a simple closed curve (Theorem 19).

What is not claimed. We do not prove that every convex body of constant width has a support function of the form we assume, so Theorem 6 is Barbier's theorem for curves *given* by such support functions, not for arbitrary convex bodies. In particular we do not prove the classical Barbier theorem for an arbitrary C^2 closed convex curve of constant width, nor for an arbitrary convex body of constant width with the perimeter of its boundary; our perimeter is the integral of the speed of γ_h, and it is identified with Mathlib's length only for smooth support functions and for the Reuleaux triangle. The Reuleaux triangle is the curve of an explicit support function: we do not prove that its body is the intersection of three discs, nor its interior angles or its area. The four-vertex theorem is proved for ovals given by C^3 support functions with $h + h'' > 0$, not for arbitrary C^2 ovals (whose support function need only be C^2) and not for general simple closed curves (Kneser's theorem). The comparison with textbook statements in Section 9.1 classifies the classical Barbier, Reuleaux and four-vertex statements as covered only in this narrower form. Section 10 lists these and the other limitations.

1.1 Availability

The proofs are in two Lean 4 developments, `barbier` (ten files, 2,746 lines) and `fourvertex` (six files, 858 lines), on Lean v4.33.1 [9] with Mathlib [10] pinned at revision `0df444a3`. They are published as `https://github.com/jpbald93/barbier` (commit `7baae04`) and `https://github.com/jpbald93/fourvertex` (commit `14d1044`); every source reference in this paper links to those commits. Version 2 of this paper adds, in a new commit of each repository (`barbier`: `d2d6d48d509a3c1d4c5e87faee319760dba31089`; `fourvertex`: `8d35e5df197fc6f416edc2b5dd6509023c5ab203`), the Layer-14 and Layer-15 files of Section 9.1: `Barbier/L14.lean` (290 lines) and `Barbier/L15Classical.lean` (578 lines), `FourVertex/L14.lean` (138 lines) and `FourVertex/L15Classical.lean` (727 lines). The pre-existing library modules and their statements are unchanged (checked by hash); the root import files and gate lists are extended, the READMEs are updated, and the tracked `scratch/Ax.lean` files are removed. Each development has a gate script: it rejects `sorry`, `admit`, `native_decide`, `axiom` and meta-programming commands in the sources, builds the library, and checks with `#print axioms` that every listed declaration (62 in `barbier` and 48 in `fourvertex` at the version-1 commits; 96 and 66 in version 2) depends only on `propext`, `Classical.choice` and `Quot.sound`. Every declaration named in this paper is in those lists (Section 9). Supporting material (the Layer-14 and Layer-15 reports, the blind writer's statements and report, the transcript audit, and the reviews of this version with the fixes made in response) is in the folder `PaperC-barbier-four-vertex` of the repository `https://github.com/jpbald93/LEAN-verification`.

2 The support-function model

We identify the plane with $\mathbb{C}$. For $h : \mathbb{R} \to \mathbb{R}$ write h' for its derivative, with the usual Lean convention that the derivative is 0 where it does not exist.

Definition 1 (curve and perimeter)**.** The *curve* of h is $\gamma_h(\theta) = (h(\theta) + i\,h'(\theta))\,e^{i\theta}$, and its *perimeter* is $P(h) = \int_0^{2\pi} |\gamma_h'(\theta)|\,d\theta$ (`Barbier.curve`, `Basic`:28; `Barbier.perimeter`, `Basic`:31).

The geometry behind the formula: $\langle \gamma_h(\theta), e^{i\theta} \rangle = h(\theta)$, so $\gamma_h(\theta)$ lies on the line with outer normal $e^{i\theta}$ at signed distance $h(\theta)$ from the origin, and when h is C^2 the velocity is $\gamma_h'(\theta) = (h + h'')(\theta) \cdot i e^{i\theta}$, tangent to that line (`Barbier.bridge_curve`, `Bridge`:37). So $\rho = h + h''$ is the radius of curvature, and the curve is convex where $\rho \geq 0$.

Definition 2 (smooth constant-width support function)**.** h is a *smooth constant-width support function of width* w (`Barbier.ConstWidthSupport`, `Basic`:21) if h is C^2 and 2π-periodic, $h(\theta) + h(\theta + \pi) = w$ for all θ, and $h + h'' \geq 0$ everywhere.

Definition 3 (piecewise constant-width support function)**.** h is a *piecewise constant-width support function of width* w (`Barbier.PiecewiseConstWidth`, `Piecewise`:23) if h is C^1 and 2π-periodic, $h(\theta) + h(\theta + \pi) = w$ for all θ, there is a countable set S such that for every $\theta \notin S$ the function h' is differentiable at θ and $h(\theta) + h''(\theta) \geq 0$, and $h + h''$ is integrable on $[0, 2\pi]$.

Countability of S is exactly what the fundamental theorem of calculus with a countable exceptional set needs (Mathlib's `intervalIntegral.integral_eq_of_hasDerivAt_off_countable_of_le`). It allows finitely many corners per period, as in the Reuleaux triangle, and more.

3 Barbier's theorem

Theorem 4 (Barbier, smooth)**.** *If h is a smooth constant-width support function of width w, then the perimeter $P(h) = \int_0^{2\pi} |\gamma_h'|$ of the curve γ_h is πw (`Barbier.barbier`, `Basic:110`).*

Proof sketch. Since $h + h'' \geq 0$, $|\gamma_h'| = h + h''$ (`Barbier.norm_deriv_curve_of_convex`, `Basic`:80). Periodicity of h' gives $\int_0^{2\pi} h'' = 0$, so $P(h) = \int_0^{2\pi} h$ (`Barbier.perimeter_eq_integral`, `Basic`:99). Splitting $[0, 2\pi]$ at π and using $h(\theta) + h(\theta + \pi) = w$ gives $\int_0^{2\pi} h = \pi w$ (`Barbier.integral_support_eq`, `Basic`:34); this step needs only continuity of h. □

Proposition 5. *Every smooth constant-width support function of width w is a piecewise one of width w (`Barbier.ConstWidthSupport.toPiecewise`, `Piecewise:57`).*

Theorem 6 (Barbier, with corners)**.** *If h is a piecewise constant-width support function of width w, then the perimeter $P(h) = \int_0^{2\pi} |\gamma_h'|$ of the curve γ_h is πw (`Barbier.barbier_piecewise`, `Piecewise:72`).*

Proof sketch. Off S the derivative of γ_h is still $(h + h'')\, ie^{i\theta}$, which needs only h to be C^1 and h' differentiable at that point, so $|\gamma_h'| = h + h''$ off S. A countable set is Lebesgue-null, so $P(h) = \int_0^{2\pi} (h + h'')$. The function h' is continuous, periodic, and differentiable off S with integrable derivative, so the countable-exception fundamental theorem of calculus gives $\int_0^{2\pi} h'' = h'(2\pi) - h'(0) = 0$. The last step is as before. □

The care needed is in the junk values: where a derivative does not exist, Lean assigns it the value 0, so on S both h'' and γ_h' may take junk values, and the proof never uses those values except through null sets.

Theorems 4 and 6 are about curves *given* by a support function, with the perimeter defined as the integral of the speed. The classical theorem for an arbitrary C^2 convex curve or an arbitrary convex body of constant width would also need that every such curve or body arises from a support function of this kind; that is not proved (Section 10).

Examples. The circle of diameter w (`Barbier.circle_perimeter`, `Examples`:23) and the smooth family $h(\theta) = a + b\cos 3\theta$ with $0 \leq 8b \leq a$, non-circular when $b > 0$ (`Barbier.trefoil_perimeter`, `Examples`:66) are instances of Theorem 4. For the circle, the curve is Mathlib's `circleMap` and the perimeter is $2\pi|r|$ (`Barbier.bridge_perimeter`, `Bridge`:122).

4 The Reuleaux triangle

Definition 7 (Reuleaux support function)**.** For $w \in \mathbb{R}$ let

$$g_w(p) = \begin{cases} \dfrac{w}{\sqrt{3}} \cos p, & p \leq \pi/6, \\ w - \dfrac{w}{\sqrt{3}} \cos(p - \pi/3), & p > \pi/6, \end{cases}$$

and let $R_w(\theta) = g_w(p)$, where p is the unique number in $[-\pi/6, -\pi/6 + 2\pi/3)$ congruent to θ modulo $2\pi/3$ (`Barbier.reuleauxPiece`, `Reuleaux`:10; `Barbier.reuleaux`, `Reuleaux`:16).

On the first branch R_w is the support function of the single corner at distance $w/\sqrt{3}$ from the centre; on the second it is the support function of the arc of radius w centred at the opposite corner. So $h + h''$ is 0 on the corner pieces and w on the arc pieces, and it jumps at six points per period.

Theorem 8 (Reuleaux triangle)**.** *Let $w > 0$. Then:*

(a) R_w is a piecewise constant-width support function of width w (`Barbier.reuleaux_constWidth`, `Reuleaux:1074`);

(b) $P(R_w) = \pi w$ (`Barbier.reuleaux_perimeter`, `Reuleaux:1083`);

(c) R_w is not twice continuously differentiable (`Barbier.reuleaux_not_smooth`, `Reuleaux:1086`);

(d) the points $\gamma(0)$, $\gamma(2\pi/3)$, $\gamma(4\pi/3)$ of the curve of R_w are pairwise at distance w (`Barbier.reuleaux_vertices`, `Reuleaux:1133`);

(e) for every $\theta \in [5\pi/6, 7\pi/6]$, $|\gamma(\theta) - \gamma(0)| = w$ (`Barbier.reuleaux_arc`, `Reuleaux:1368`);

(f) the total variation of γ_{R_w} on $[0, 2\pi]$ is finite (`Barbier.reuleaux_length_ne_top`, `ReuleauxLength:296`), equals our perimeter (`Barbier.reuleaux_perimeter_eq_eVariationOn`, `ReuleauxLength:300`), and equals πw (`Barbier.reuleaux_length`, `ReuleauxLength:248`).

Part (b) is Theorem 6 applied to (a). Part (c) is what makes (b) new: by (c), R_w satisfies no C^2 hypothesis, so the smooth Theorem 4 cannot be applied to it. The proof of (c) computes the one-sided limits of R_w'' at $\pi/6$, namely $-w/2$ and $w/2$, and contradicts continuity of the second derivative. Parts (d) and (e) establish an equilateral triple of points of side w on the curve and, on one parameter interval, that the curve lies at distance w from the opposite point (rotated counterparts follow by the $2\pi/3$-periodicity of R_w); they do not establish that the trace of the curve equals the boundary of the intersection of the three discs. We do not prove that the body of R_w (Section 6) is the intersection of the three closed discs of radius w about the corners, nor the interior angle $2\pi/3$ at each corner, nor the area $\frac{1}{2}(\pi - \sqrt{3})w^2$.

Most of the 1,379 lines of `Reuleaux.lean` are in (a): C^1-smoothness across the wrap-around points of the angle reduction, where one-sided derivatives must be matched, and the bookkeeping of the reduction modulo $2\pi/3$ (Mathlib's `toIcoMod`).

5 The four-vertex theorem

The `fourvertex` development uses the same curve γ_h (`FourVertex.curve`, `Support`:25) and the radius of curvature $\rho = h + h''$ (`FourVertex.radius`, `Support`:17).

Definition 9. A function g has *four sign changes* (`FourVertex.FourSignChanges`, `Basic`:18) if there are $t_1 < t_2 < t_3 < t_4 < t_1 + 2\pi$ with $g(t_1) > 0$, $g(t_2) < 0$, $g(t_3) > 0$, $g(t_4) < 0$. A function f has *four vertices* (`FourVertex.FourVertices`, `Vertices`:73) if there are $c_1 < c_2 < c_3 < c_4 < c_1 + 2\pi$ at which f has, in order, a local maximum, minimum, maximum and minimum.

Lemma 10 (Sturm–Hurwitz type)**.** *Let g be continuous and 2π-periodic with $\int_0^{2\pi} g = \int_0^{2\pi} g\cos = \int_0^{2\pi} g\sin = 0$. Then $g \equiv 0$ or g has four sign changes (`FourVertex.eq_zero_or_fourSignChanges`, `Basic:150`).*

Proof sketch. If g has at most two sign changes, choose a, b with $\cos(t - a) - \cos b$ having the same sign pattern as g; then $g(t)(\cos(t - a) - \cos b) \geq 0$ and is not identically 0, but its integral is 0 by the three orthogonality conditions, a contradiction. Sign changes come in pairs by periodicity. □

Theorem 11 (four-vertex, radius form)**.** *If h is C^3 and 2π-periodic, then ρ is constant or ρ' has four sign changes (`FourVertex.radius_const_or_fourSignChanges`, `Support:124`).*

Integration by parts shows ρ' is orthogonal to 1, cos and sin (the terms $h' + h'''$ integrate against cos and sin to 0), so Lemma 10 applies. No convexity is needed for this analytic statement.

Theorem 12 (four-vertex theorem for support-function ovals)**.** *Let h be C^3 and 2π-periodic with $\rho > 0$ everywhere. Then the curvature $1/\rho$ is constant, or there are $c_1 < c_2 < c_3 < c_4 < c_1 + 2\pi$ at which $1/\rho$ has, in order, a local minimum, maximum, minimum and maximum (`FourVertex.four_vertex`, `Vertices:103`).*

The step from four sign changes of ρ' to four alternating extrema of ρ is `FourVertex.fourVertices_of_fourSignChanges`; taking reciprocals swaps maxima and minima. The oval with support function $5 + \cos 2\theta$ is a non-circular example (`FourVertex.oval`, `Examples`:11; `FourVertex.oval_four_vertex`, `Examples`:51): its radius of curvature ranges over $[2, 8]$, attaining both ends (`FourVertex.bridge_oval`, `Bridge`:111).

6 Bridges to standard notions

The theorems above are stated with our own definitions: our perimeter, our constant-width condition, our curvature. This section proves that those definitions mean what they are meant to.

6.1 Length

Mathlib defines the length of a curve on $[a, b]$ as its total variation `eVariationOn`: the supremum over partitions of $\sum |\gamma(t_{k+1}) - \gamma(t_k)|$, a value in $[0, \infty]$. At our Mathlib pin there is no lemma computing it as an integral.

Theorem 13 (length of a C^1 curve)**.** *If $\gamma : \mathbb{R} \to \mathbb{C}$ is C^1 and $a \le b$, then the total variation of γ on $[a, b]$ is finite and equals $\int_a^b |\gamma'(t)|\, dt$ (`Barbier.length_eq_integral_norm_deriv`, `ArclenFinite:50`; the finiteness alone is `Barbier.eVariationOn_ne_top_of_contDiff`, `ArclenFinite:7`, and the equation alone `Barbier.eVariationOn_eq_integral_norm_deriv`, `Bridge:144`).*

Proof sketch. For a partition, $|\gamma(t_{k+1}) - \gamma(t_k)| \le \int_{t_k}^{t_{k+1}} |\gamma'|$; summing gives variation $\le \int_a^b |\gamma'|$, in particular finiteness. For the converse, let $V(s)$ be the variation on $[a, s]$. Additivity of variation over adjacent intervals and the two-sided estimate $|\gamma(s+\delta) - \gamma(s)| \le V(s+\delta) - V(s) \le \int_s^{s+\delta} |\gamma'|$, both $\delta |\gamma'(s)| + o(\delta)$ by uniform continuity of γ', give $V'(s) = |\gamma'(s)|$ from the right, and the fundamental theorem of calculus concludes. □

The finiteness clause matters in Lean: the conversion of $[0, \infty]$ to $\mathbb{R}$ sends ∞ to 0, so the equation alone would not by itself exclude infinite length. It was added after the blind check of Section 7 flagged the difference.

Corollary 14 (perimeter is length)**.** *If h is C^2 and 2π-periodic with $h + h'' \ge 0$, then $P(h)$ is the total variation of γ_h on $[0, 2\pi]$ (`Barbier.perimeter_eq_eVariationOn`, `Bridge:165`).*

For the Reuleaux triangle γ' is discontinuous, so Corollary 14 does not apply to it as stated; the Reuleaux triangle is covered separately by Theorem 8(f), and a general perimeter-equals-length bridge for the whole piecewise class is left open (Section 10).

6.2 The convex set

Mathlib has no support function (its convex-body file lists support sets as future work), so we construct the set directly.

Definition 15. The *body* of h is $K_h = \bigcap_\theta \{z : \mathrm{Re}(z\,e^{-i\theta}) \le h(\theta)\}$ (`Barbier.body`, `Bridge`:172).

Theorem 16 (the body)**.** *K_h is convex for every h (`Barbier.convex_body`, `Bridge:175`). If h is a piecewise constant-width support function of width w then, for every θ:*

(a) $h(\theta)$ is the greatest value of $\langle \gamma_h(\varphi), e^{i\theta}\rangle$ over $\varphi \in \mathbb{R}$, attained at $\varphi = \theta$ (`Barbier.bridge_support_piecewise`, `BridgePiecewise:109`);

(b) $\gamma_h(\theta) \in K_h$ (`Barbier.curve_mem_body_piecewise`, `BridgePiecewise:170`);

(c) the supremum of $\mathrm{Re}(z\,e^{-i\theta})$ over $z \in K_h$ is $h(\theta)$ (`Barbier.support_body_piecewise`, `BridgePiecewise:177`);

(d) the support values of K_h in directions θ and $\theta + \pi$ add up to w (`Barbier.width_body_piecewise`, `BridgePiecewise:184`).

The smooth versions are `Barbier.curve_mem_body`, `Barbier.support_body` and `Barbier.width_body` (`Bridge:180–197`).

Theorem 17 (the body is a convex body)**.** *If h is a piecewise constant-width support function of width w, then K_h is compact (`Barbier.isCompact_body_piecewise`, `ConvexBodyBridge:57`) and nonempty (`Barbier.body_nonempty_piecewise`, `ConvexBodyBridge:60`); with convexity this makes K_h a Mathlib `ConvexBody` whose support values are h and whose width in every direction is w (`Barbier.exists_convexBody_piecewise`, `ConvexBodyBridge:64`). In particular, for $w > 0$, the Reuleaux triangle is a Mathlib convex body of constant width w (`Barbier.reuleaux_convexBody`, `ConvexBodyBridge:75`).*

Boundedness is the only new input: h is continuous and periodic, so $|h|$ is bounded by M; a point z of K_h is bounded by the constraint in the direction of z itself, $\|z\| = \langle z, e^{i\arg z}\rangle \le h(\arg z) \le M$.

So h is the support function of the convex set K_h, and K_h has width w in every direction: it is a set of constant width in the standard sense, and by Theorem 8 this includes the body of the Reuleaux support function R_w. The proof of (a) differentiates $\varphi \mapsto \langle \gamma_h(\varphi), e^{i\theta}\rangle$, with derivative $\rho(\varphi)\sin(\theta - \varphi)$ off S, and integrates. A supremum of an unbounded set of reals is 0 in Lean, so (c) and (d) are proved through attainment (part (a) and part (b)), not read off a supremum that might be junk.

6.3 Curvature and simplicity

Mathlib has no curvature of plane curves, so we define the textbook one.

Definition 18. The *signed curvature* of $\gamma : \mathbb{R} \to \mathbb{C}$ at t is $\kappa_\gamma(t) = \mathrm{Im}(\overline{\gamma'(t)}\,\gamma''(t))/|\gamma'(t)|^3$, which is $(x'y'' - y'x'')/(x'^2 + y'^2)^{3/2}$ for $\gamma = x + iy$ (`FourVertex.signedCurvature`, `Curvature`:15).

Theorem 19 (curvature and simplicity)**.** *(a) If h is C^3 and $\rho(\theta) > 0$, then $\kappa_{\gamma_h}(\theta) = 1/\rho(\theta)$ (`FourVertex.signedCurvature_curve`, `Curvature:129`).*

(b) If h is 2π-periodic, so is γ_h (`FourVertex.curve_periodic`, `Curvature:168`).

(c) *If h is C^3 and 2π-periodic with $\rho > 0$ everywhere, then γ_h is injective on $[0, 2\pi)$ (`FourVertex.curve_injOn`, `Curvature:179`).*

For (a), $\gamma_h' = \rho\, i e^{i\theta}$ (`FourVertex.bridge_radius`, `Bridge`:38) gives $\gamma_h'' = \rho'\, i e^{i\theta} - \rho\, e^{i\theta}$, so $\overline{\gamma_h'}\gamma_h'' = \rho\rho' + i\rho^2$ and $\kappa = \rho^2/\rho^3$. For (c), the function $\varphi \mapsto \langle \gamma_h(\varphi), e^{i\theta} \rangle$ has derivative $\rho(\varphi)\sin(\theta - \varphi)$, so it is strictly increasing on $[\theta - \pi, \theta]$ and strictly decreasing on $[\theta, \theta + \pi]$, with maximum $h(\theta)$ attained only at θ in that window. If $\gamma_h(a) = \gamma_h(b)$ with $a \neq b$ in $[0, 2\pi)$, reduce b into the window around a by periodicity and contradict strictness.

So under the hypotheses of Theorem 12, γ_h is a simple closed curve and $1/\rho$ is its signed curvature. The theorem can therefore be restated with no reference to support functions in its conclusion:

Theorem 20 (four-vertex theorem, textbook form)**.** *Let h be C^3 and 2π-periodic with $\rho > 0$. Then the signed curvature of the simple closed curve γ_h is constant, or it has, at some $c_1 < c_2 < c_3 < c_4 < c_1 + 2\pi$, a local minimum, maximum, minimum and maximum, in order (`FourVertex.four_vertex_signedCurvature`, `Curvature:228`).*

The textbook form is about the curves γ_h of such support functions. An arbitrary C^2 oval is not covered: its support function need only be C^2, so ρ need not be differentiable and the argument through ρ' does not apply, and we do not prove that every C^3 oval arises from a C^3 support function.

The four points are distinct points of the circle $\mathbb{R}/2\pi\mathbb{Z}$, and each is a local extremum in Mathlib's sense (`FourVertex.bridge_FourVertices`, `Bridge`:95; `FourVertex.bridge_FourSignChanges`, `Bridge`:64).

7 Does the statement mean what it says?

A complete proof only shows that *the Lean statement* is true. To test whether the Lean statements say what this paper says, we ran a blind re-formalisation in three rounds.

Round 1 (before this paper). Fifteen `barbier` results were written out in plain language by the author's assistant from the code, translated into Lean by an independent model, and compared. Fourteen pairs were equivalent at once. The fifteenth was the length lemma: the blind statement asserted finiteness and ours did not. We strengthened ours (Theorem 13 now states finiteness), after which that pair was equivalent too.

Round 2 (this paper). The mathematical sections of this draft, with every code name, source reference and description of the formalisation removed automatically, were given to a model different from the one that wrote the proofs. It produced Lean statements of all 22 results and parts of Sections 3–6, reading only that text (self-reported; Mathlib API lookups allowed). A first attempt was rejected on review because two of its definitions were not faithful (length defined as the constant 0, and the Reuleaux angle reduction omitted); the second attempt was reviewed definition by definition. A third model then proved, for each pair, that the blind statement and ours imply each other, using only unfolding of definitions, repackaging and general Mathlib lemmas, never one of our theorems. The Reuleaux-length and convex-body results were added to the draft afterwards and checked the same way in a third round, with seven further blind statements (the only difference being a form one, ofReal against toReal, in the pair for Theorem 8(f)).

Result. Twenty-six of the 29 pairs are equivalent in both directions, including the Reuleaux support function (which the blind writer defined with a floor function rather than `toIcoMod`), the piecewise hypothesis class, the body and its support values, and both forms of the four-vertex theorem. A twenty-seventh pair, the Reuleaux perimeter, is equivalent in both directions once the finiteness of the length (pair 23) is supplied as a separate hypothesis. The other two pairs differed in one direction only, and in the safe one: an earlier version of this draft stated Theorem 19(a) and (b) under the hypotheses of part (c), while the Lean theorems need fewer (no periodicity for (a); no smoothness for (b)). Our statements were therefore stronger than claimed. We reworded Theorem 19 so each part states exactly its own hypotheses.

Caveats. The blind writers self-reported which files they read. The comparison establishes equivalence of statements, not correctness of the prose proofs.

8 Related work

We searched for prior formal proofs of both theorems in Mathlib (pinned and current), its pull requests and issues, the Archive of Formal Proofs [11], Coq libraries (GeoCoq, MathComp and its analysis library), HOL Light, HOL4, Mizar, Metamath (`set.mm` and `iset.mm`), the "100 theorems" and "1000+ theorems" lists, GitHub code search, the public Lean Zulip archive, and several benchmark and autoformalisation collections. We count only a complete, sorry-free proof as prior art.

Barbier. We found no formal statement or proof in any system, and no formalisation of the Reuleaux triangle. The "1000+" entry for Barbier's theorem is empty in every system column.

Four-vertex. The closest work is in `facebookresearch/atlas-lean` [12] (file `FourVertex.lean`, present from the repository's initial commit of 28 May 2026), which states an incomplete four-vertex theorem for smooth positive-curvature parametrised simple closed curves, concluding four stationary points of the curvature rather than the alternating local extrema used here. Its Sturm–Hurwitz lemma and its final assembly step are `sorry`. We compiled it at our Mathlib pin (after eight line-local compatibility edits, no statement changed), and `#print axioms` confirms that its four-vertex theorem and its Sturm–Hurwitz lemma depend on `sorryAx`. A second Lean repository (`mirajcs/TheFourVertexTheorem`) [13] has definitions and `sorry`s but no four-vertex statement. A 2025 Zulip thread records an intention to formalise the theorem, with no code posted. To our knowledge, Lemma 10 and Theorem 12 are the first complete formal proofs of a Sturm–Hurwitz-type lemma and of a four-vertex theorem.

Support functions. Mathlib has convex bodies (`ConvexBody`) but no support function. The nearest external Lean work is `deancureton/MovingSofa` [14] (September 2026), an autoformalisation of Baek's [15] moving-sofa proof, which defines support values and directional width of a set and uses `eVariationOn` for boundary length. It states neither theorem treated here.

Our earlier papers. Two earlier papers in this series formalised classical Euclidean results by coordinate certificates: Paper A [16] and Paper B [17]. The present paper is the first in the series on analytic geometry of curves.

9 Verification

Gates. Each repository's gate script is described in Section 1.1. Both gates passed on two machines of different architecture, an x86-64 Linux machine and an arm64 macOS machine: `barbier` with 62 declarations and `fourvertex` with 48, all with only the three standard axioms. The runs were made on 9–10 October 2026 on the sources that were subsequently published; the published commits contain byte-identical sources (checked by hash), so the gate results apply to them. With the Layer-14 and Layer-15 files of Section 9.1 (version 2), the gates pass with 96 and 66 declarations, again with only the three standard axioms, and both tampering suites below were re-run and rejected every planted defect (runs of 21–22 October 2026, on the arm64 machine).

Tampering tests. Each repository has a tampering suite that copies the project, plants one defect and checks that the gate rejects it at the expected stage: a hidden `sorry`, an `axiom` declaration split across two lines, a fake `#print axioms` report printed by `#eval`, a macro that could redefine commands, a dropped hypothesis that breaks a proof, and an extra axiom that only the output parser can catch. All six were rejected in both repositories (run of 10 October 2026, on the x86-64 machine).

Statements not weakened. The Reuleaux and corner statements were written first, as statements with `sorry` proofs, then proved by coding agents under the rule that no statement may change. We compared every statement and definition before and after: all are word-for-word identical. The pre-existing smooth Barbier and four-vertex theorem statements and proofs are unchanged by the work of this paper; the bridge modules and the root import files were extended with new lemmas and imports, while the pre-existing `Barbier/Basic.lean` and `Barbier/Examples.lean` are byte-for-byte unchanged from their earlier versions.

Statements checked by hand. The author read every statement quoted in this paper against the corresponding Lean source by hand, and confirmed that each informal statement agrees with the formal statement it quotes.

Statement-level checks. Beyond the gate and the tampering suite, both developments were run through a statement-level battery. It mutates each theorem (numeric literals, inequalities and equalities altered) and checks that the proof rejects the mutant; it probes the hypotheses for consistency; it tries to certify each statement satisfiable; it ablates each hypothesis to see whether the proof uses it; and it lints for unused variables. Results: `barbier` 386 mutants, 370 rejected, 16 survived; `fourvertex` 91 mutants, 86 rejected, 5 survived. Every survivor is a mutant that is still *true* — a looser numeric bound (e.g. $2w$ to $3w$), a constant changed in an auxiliary inequality that stays valid ($\pi/6$ to $\pi/7$), or a strictly stronger hypothesis — so survival is expected and is not a defect: the proofs simply do not depend on the exact constant. The contradiction probe found no contradictory hypothesis set. Satisfiability was certified for 101 of the 168 `barbier` declarations and 26 of the 47 `fourvertex` declarations; the rest are recorded as unresolved rather than assumed, because no witness was produced. Hypothesis ablation found every hypothesis used except two in `barbier` (`Barbier.perimeter_eq_eVariationOn` does not use its convexity hypothesis, and one auxiliary interval lemma does not use its hypothesis $w > 0$), both of which make the theorem slightly more restrictive than necessary without affecting its correctness. The unused-variable linter reports no warnings.

Name gate. A script checks that every declaration named in this paper is in its repository's gate list and that every source anchor points at a line declaring that name.

9.1 Layers 14 and 15

Layer 14: statements in Mathlib's language and non-vacuity. Each of the 13 main results (Theorems 4, 6, 8, 13, 16 and 17, Proposition 5 and Corollary 14 in `barbier`; Lemma 10 and Theorems 11, 12, 19 and 20 in `fourvertex`) is restated in a file `Barbier/L14.lean`, respectively `FourVertex/L14.lean`, with Mathlib-only types, and proved from the library theorem through bridges that hold by `rfl` or structure eta. The curve, perimeter, body, curvature and sign pattern are written out (e.g. ρ as $h + h''$, the signed curvature as $\mathrm{Im}(\overline{\gamma'}\gamma'')/|\gamma'|^3$, the body as an intersection of half-planes, the Reuleaux function by its defining two-branch equation with `toIcoMod`). A mechanical check of each elaborated type found no constant of either library (12 `barbier` and 7 `fourvertex` restatements and witnesses). Theorem 13 was already Mathlib-only and is re-exported. Three restatements say slightly more: the smooth Barbier theorem also gives Mathlib length πw (`Barbier.l14_barbier`), the perimeter-is-length corollary also states finiteness (`Barbier.l14_perimeter_eq_eVariationOn`), and the textbook four-vertex theorem includes injectivity on $[0, 2\pi)$ (`FourVertex.l14_four_vertex_signedCurvature`). Every hypothesis set has an explicit non-degenerate witness:

witness	used for; non-degeneracy proved
$h = 2 + \cos(3\theta)/4$, $w = 4$ (`Barbier.l14w_barbier`)	smooth Barbier, Prop. 5, Cor. 14; $h(0) \neq h(\pi/3)$, $\gamma(0) \neq \gamma(\pi)$
R_1 (`Barbier.l14w_reuleaux`)	Barbier with corners, Reuleaux, body; not C^2, corners pairwise distinct
$\gamma(t) = t$ on $[0, 1]$ (`Barbier.l14w_length`)	length; $\gamma(0) \neq \gamma(1)$, integral 1
$g = 6 \sin 2t$ (`FourVertex.l14w_sturm`)	Sturm–Hurwitz; $g(\pi/4) = 6 \neq 0$
$h = 5 + \cos 2\theta$ (`FourVertex.l14w_oval`)	four-vertex forms, curvature; $\kappa(0) \neq \kappa(\pi/2)$, four ordered extrema

No Mathlib-only restatement turned out false or needed an extra hypothesis. Helper, bridge and example declarations named in the paper (such as `FourVertex.bridge_FourVertices` or `Barbier.trefoil_perimeter`) were not restated, since they are not main results. One observation: for the piecewise class (Theorem 6, Theorem 8(b)) the Mathlib-only statement is about $\int \|\mathrm{deriv}\, \gamma\|$, with Lean's junk value 0 for the derivative where it does not exist, not about Mathlib's length; this is the limitation already stated in Section 10.

Layer 15: comparison with the classical statements. Another AI agent collected the classical statements from the literature, with citations (Barbier [1] and the textbook forms in Lay and do Carmo; the Reuleaux triangle; Mukhopadhyaya [2], Kneser [3] and Osserman [5]): C1 Barbier in a smooth form for a C^2 closed convex curve, a support-function form and a general form for any convex body of constant width; C2 the Reuleaux triangle as an intersection of three discs (boundary, constant width, corners and angles, perimeter, area and Blaschke–Lebesgue, Reuleaux polygons); C3 the four-vertex theorem for C^2 ovals (with a more precise form and do Carmo's variant), Kneser's form, and the remarks on the ellipse, the circle and non-simple curves. A writer model, given only these statements and no code or paper, wrote 17 Lean statements with Mathlib notions and its own definitions (plane convex body, support function, width, perimeter as a supremum of variations over boundary parametrisations, signed curvature, vertex as a local extremum), marked seven ambiguities and no extra hypothesis, and separated two well-formedness obligations; its transcript was audited (a transcript audit in the verification repository found no read of any file

other than the writer's own brief, outputs and compiler output, and no web access). A comparer, copying the writer's statements verbatim and transporting $\mathbb{C}$ to $\mathbb{R}^2$ by Mathlib's isometry, proved each statement from the development or classified it:

- *Covered* (3): the support-function form of Barbier (`Barbier.Classical.cov_C1_a_support`; the writer asks $h + h'' > 0$, stronger than our ≥ 0), and two remarks that the paper does not claim, the circle (`FourVertex.Classical.cov_C3_remark_circle`) and the ellipse with $a \neq b$, whose vertices are exactly $0, \pi/2, \pi, 3\pi/2$ (`FourVertex.Classical.cov_C3_remark_ellipse`).

- *Narrower* (9): the development proves the statement only for curves given by support functions. Barbier, smooth form (`Barbier.Classical.cov_C1_a_part`): missing that every C^2 convex curve of constant width is a reparametrisation of some γ_h, and reparametrisation invariance. Barbier, general form and its equivalent form (`Barbier.Classical.cov_C1_b_part`, `Barbier.Classical.cov_C1_b_equiv_part`): missing that every constant-width body is some K_h with h as in Definition 3, that the boundary of K_h is the trace of γ_h with perimeter equal to its variation, and perimeter equals length in the piecewise class; nonempty interior of K_h is not proved. Reuleaux boundary, constant width and corners, perimeter (`Barbier.Classical.cov_C2_boundary_part`, `Barbier.Classical.cov_C2_a_part`, `Barbier.Classical.cov_C2_b_part`): missing the identification of K_{R_w} or γ_{R_w} with the intersection of three discs, and the interior angles. Four-vertex for ovals, its precise form and do Carmo's variant (`FourVertex.Classical.cov_C3_a_part`, `FourVertex.Classical.cov_C3_a_precise_part`, `FourVertex.Classical.cov_C3_a_doCarmo_part`): missing that every oval is γ_h for a C^3 support function with $\rho > 0$ (false in general for a C^2 oval, whose ρ need not be differentiable), and reparametrisation invariance of vertices. For C^3 2π-periodic support functions with $\rho > 0$, the transported curve γ_h satisfies the writer's hypotheses of the C^2 four-vertex statement, and its conclusion (four increasing vertices in $[0, 2\pi)$) is proved. For do Carmo's variant only the conclusion (four increasing stationary points of the curvature in $[0, 2\pi)$) is proved; C^3 regularity of the curve is not established. For a piecewise constant-width support function of width $w > 0$, K_h has constant width w in the writer's sense.

- *Out of scope* (5): the area of the Reuleaux triangle; Blaschke–Lebesgue (only the disc area $\pi w^2/4$ is proved, `Barbier.Classical.cov_C2_c_disk_area`); Reuleaux polygons (only that the intersection of the discs is a convex body, `Barbier.Classical.cov_C2_reuleaux_polygon_part`); Kneser's theorem for non-convex simple curves; and the non-simple curve with two vertices.

- *Writer statement false as written*: none. No statement was shown false.

New proofs, not in the main files: `Barbier.Classical.cov_C1_a_support`, `FourVertex.Classical.cov_C3_remark_circle`, `FourVertex.Classical.cov_C3_remark_ellipse`; the obligation that a convex body attains its support value (`Barbier.Classical.cov_supportFn_obligation`, `FourVertex.Classical.cov_supportFn_obligation`); that K_h has constant width in the writer's sense (inside `Barbier.Classical.cov_C1_b_part`); that an intersection of a nonempty finite family of closed discs of common radius $w > 0$ about points pairwise at distance at most w is a plane convex body (`Barbier.Classical.isPlaneConvexBody_iInter`, `Barbier.Classical.isPlaneConvexBody_reuleaux`), and the writer's Reuleaux triangle is not a disc (`Barbier.Classical.reuleaux_not_disk`); the angles $\pi/3$ of the equilateral triangle (`Barbier.Classical.angle_equilateral`); the disc area; and the transport of the four-vertex theorem to the writer's curvature and to $[0, 2\pi)$ behind the three four-vertex `_part` theorems. The

obligation that a convex body has a simple closed boundary parametrisation was not proved; no theorem depends on it. All proofs depend only on the three standard axioms. They are collected in `Barbier/L15Classical.lean` and `FourVertex/L15Classical.lean` (namespaces `Barbier.Classical` and `FourVertex.Classical`), whose gates list every comparison theorem; the writer's definitions and statements are reproduced there verbatim, and uncovered statements remain definitions of propositions. The earlier library files are byte-identical.

10 Limitations and further work

1. **Barbier for curves given by support functions.** We do not prove that every convex body of constant width has a support function satisfying Definition 3, nor that every C^2 closed convex curve of constant width is a reparametrisation of some γ_h. That would need the general theory of support functions of convex bodies (absent from Mathlib) and a regularity argument. Nor do we prove that the boundary of K_h is the trace of γ_h and has a perimeter (in the sense of inscribed polygons) equal to $P(h)$. The classical general form of Barbier's theorem is therefore covered only in the narrower form of Section 9.1.

2. **The Reuleaux triangle.** We do not identify K_{R_w} with the intersection of three closed discs, and do not prove its interior angles, its area or the Blaschke–Lebesgue theorem; general Reuleaux polygons are not treated.

3. **Four-vertex for ovals.** Theorem 20 covers strictly convex curves given by C^3 support functions with $h + h'' > 0$, not arbitrary C^2 ovals of arbitrary period. The general theorem for simple closed curves [3], and the converse [6], are not attempted.

4. **Perimeter versus length in the piecewise class.** Corollary 14 bridges our perimeter to Mathlib's length only for smooth support functions, and Theorem 8(f) for the Reuleaux triangle; we do not prove a general bridge for the whole piecewise class.

5. **Blind checks** establish equivalence of statements, not correctness of the prose proofs, and rely on the writers' self-reports (Section 7).

Acknowledgements and use of AI

A large-language-model AI assistant was used to write the Lean proofs, to run the prior-art searches and the verification battery (including Layers 14 and 15), and to draft this paper. The author directed the work and reviewed the paper. Every proof is checked by the Lean kernel, and the author takes responsibility for all content.

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